Power system flexibility: a regulator's perspective

ATHENS COURSE - KULEUVEN

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A brief introduction to the CREG

A brief introduction to the CREG: establishment and role

The **Commission for Electricity and Gas Regulation** (CREG) is a federal body (autonomous organisation granted with legal personality, established by the Electricity and Gas Laws) for the regulation of Belgian electricity and gas markets

The CREG's role results from:

- European legislation,
- transposed into Belgian laws (on 29 April 1999): leading to the CREG's foundation on **10 January 2000**

and was subsequently strengthened and refined through, among others:

- **3rd Energy Package** (2009): strengthening regulatory authories' powers and independence;
- **Clean Energy Package** (2019)
- **RePowerEU Package**

These legislative packages allocated new tasks and responsibilities to regulatory authorities (regarding to, among others, consumer rights, dynamic prices, price comparison tools, reports on TSO tariffs, hydrogen regulation, etc.)

(NRA = National Regulatory Authority ; TSO = Transmission System Operator)

more info on https://www.creg.be/en/presentation-creg

A brief introduction to the CREG: mission

The CREG acts **completely independent** from governments and other stakeholders *in accordance with one of the cornerstones of the European liberalization process*

link to report on NRA independence

Its main mission is twofold:

- enabling & supervising the **functioning of electricity, gas and hydrogen markets**
- protecting the **interest of consumers**

It does so by:

- advising public authorities
- monitoring transparency and competition on the markets
- deciding and enforcing decisions in accordance with the applicable legislation
- informing and empowering all consumers: vulnerable, small and medium-sized enterprises as well as industrial consumers

A brief introduction to the CREG: organisation

The CREG's staff consists of ± 80 employees, led by the Board of Directors
(1 President of the Board of Directors and three Directors, one for each Directorate)

V **Presidency**, led by Mr. Koen Locquet

A **Administrative Directorate**, led by Ms. Ilse Tant

T **Technical Operation of Electricity and Gas Markets Directorate**, led by Mr. Laurent Jacquet

P **Prices and Accounts Control on the Electricity and Gas Markets Directorate**, led by Ms. Sigrid Jourdain

Adviesraad / Conseil Consultatif is an advisory council and stakeholder forum with representatives from CREG, governments (federal and regional), sector representatives (consumers, suppliers, producers, etc.),…

The CREG reports to, is answerable to and receives its budget from the **federal parlement**
(Kamer van volksvertegenwoordigers / Chambre des représentants)

A brief introduction to the CREG: work at the European level

The **Council for European Energy Regulators** (CEER)

- 👉 is a voluntary, not-for-profit organisation to support NRAs
- 👉 supported by a Secretariat
- 👉 based in Brussels, Belgium
- 👉 acts as platform for cooperation, information exchange and assistance for NRAs
- 👉 has a staff of ± 10

The **Agency for the Cooperation of Energy Regulators** (ACER)

- 👉 is an official EU body (decentralized agency)
- 👉 mandated by the Electricity and ACER Regulations
- 👉 based in Ljubljana, Slovenia
- 👉 fosters NRA coordination and is assigned with specific tasks through legislation
- 👉 has a staff of ± 100 (and growing)

ACER's most important bodies,
(in addition to permanent staff led by Director):

- **Board of Regulators**
- **Working Groups and Task Forces**
- **Board of Appeal**
- **Administrative Board**

A brief introduction to the CREG: collaboration with regional regulators

FORBEG is the forum of the Belgian electrcitiy and gas regulators: an informal consultation body consisting of representatives of BRUGEL, the CREG, CWaPE and VNR 🤝

Main goal is to **build and exchange knowledge** regarding evolutions on the (Belgian) energy markets and federal and regional policy initiatives

Different working groups exist:

- 👉 WG Electricity
- 👉 WG Gas
- 👉 WG Renewable Energy Sources
- 👉 WG Smart Meters
- 👉 WG Tariffs
- 👉 WG Information Exchange
- 👉 WG Europe

Joint deliverables, for example:

🌐 link to European price comparison study

A brief introduction to the CREG: assigned tasks and powers

- **Market regulation**: ensuring efficient functioning of wholesale electricity, gas and hydrogen markets
- **Infrastructure development**: overseeing the development and maintenance of energy infrastructure
- **Market monitoring**: supervising market transactions to ensure compliance with regulations
- **Consumer advocacy**: protecting and empowering consumers, particularly small and vulnerable ones
- **Policy advisory role**: advising public authorities on market organization and functioning
- **Enforcement**: taking actions to ensure energy undertakings' compliance with the regulatory framework
- **Collaboration**: active participation in European energy initiatives, including ACER and CEER.

link to the policy paper

A brief introduction to the CREG: division of institutional competences in Belgium

Federal competences

- Security of supply
- National prospective studies (E, G, H)
- Nuclear fuel cycles and R&D programs
- Large storage facilities
- Production and transport of electricity (> 70 kV)
- Transport tariffs, LNG terminals, storage, interconnection tariffs
- Energy statistics, energy balances, wholesale market monitoring
- Offshore wind energy
- H2

Regional competences

- Rational energy use
- New sources and renewable energies
- R&D on non-nuclear energy
- Market regulation for distribution
- Distribution and transmission of electricity (< 70kV)
- District heating equipment and networks
- Recovery of residual energy from industry or other uses
- Regional energy statistics, regional energy balances, retail market monitoring
- CO_2

Complex institutional setup implies a complex division of tasks, responsibilities and regulatory oversight!

Selected current challenges related to flexibility

Flexibility is a broad concept

Different purposes:

- **Balancing Flexibility:**
  The capability to rapidly adjust production or demand to maintain real-time balance, including services such as FCR, aFRR, and mFRR.
- **Congestion Management Flexibility:**
  Local flexibility used to relieve grid constraints, including flexible connection agreements and redispatch actions defined.
- **Adequacy Flexibility**:
  Medium- to long-term flexibility ensuring sufficient capacity to meet demand under all scenarios, highlighted in Elia's 2026-2036 adequacy and flexibility study.
- **Non-Frequency Ancillary Flexibility:**
  Flexibility related to voltage control, reactive power, or black-start services.
- …

Different ways to achieve flexibility:

Flexible assets, demand-side flexibility, …

Network development

- Required to avoid/solve congestion in the future

- Adequacy & Flexibility Study (AdeqFlex)
- Flexibiity Needs Assessment (FNA)
- Federal Development Plan: foreseen investments on the network. Formal opinion of the CREG

- **Challenges:**

  correct estimation of the needs

  consistency and alignment with the regional development plans (transport and distribution)

  Future introduction of a TOTEX-based approach: remunerated flex as an alternative to grid reinforcement

Balancing markets

- European integration
- Increasing number of flexible assets

- **Challenges:**

  Implicit vs. explicit balancing

  Definition of the imbalance price

Congestion management

- Flexible Connection Agreements (FCA): now the rule rather than the exception
- Global issue of the access to the grid: new Code of Conduct

- **Challenges:**

  Special treatment of flexible assets ?

  Activation rules (exceptions?)

  Remuneration? Incl. beyond the contractual limits

  Regulatory alignment between regulators

  Operational alignment between the TSO and the DSOs

Market surveillance

- New behaviours of flexible assets

- **Challenges:**

  Detecting strategic bidding:
  price manipulation, capacity withholding, power abuse, …

  Definition of opportunity cost:
  price spread in wholesale markets, vs. implicit balancing,
  vs. explicit balancing services

CREG

Thanks for your attention.

Commissie voor de Regulering van de Elektriciteit en het Gas